

VARSHA SATHISKUMAR

Denver, CO | (720)-255-6441 | varsha_sathiskumar@mines.edu | [linkedin.com/in/varsha-sathiskumar](https://www.linkedin.com/in/varsha-sathiskumar) | varshasathiskumar.github.io

PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building scalable full-stack enterprise and B2B web applications, with recent experience developing Agentic AI systems using RAG, LLMs, and multi-agent architectures. Experienced in taking features from design through production and delivering measurable business impact.

EDUCATION

Colorado School of Mines

Master of Science in Computer Science - GPA: 3.6/4.0

| Golden, CO

Dec 2026

Thiagarajar College of Engineering

Bachelor of Technology in Information Technology - GPA: 8.7/10.0

| Madurai, India

Apr 2021

WORK EXPERIENCE

AI Software Engineer Intern | NociQuant Inc, Minnesota City, MN

Jun 2026 – Aug 2026

- Developed a full-stack healthcare application with Python backend services integrating a 5-layer AI diagnostic pipeline, and a React/TypeScript interface for reviewing diagnostic results to support earlier chronic pain diagnosis.
- Built an end-to-end diagnostic pipeline integrating RAG, multi-agent orchestration, and an automated evaluation framework, improving retrieval recall by 20% and diagnostic accuracy from 68% to 78% across 1,000 clinical cases using Python, LangChain, LangGraph, and Pinecone.
- Automated a evaluation framework to validate 6 AI diagnostic configurations across 20+ evaluation criteria, identifying accuracy and retrieval failures through Python-based test automation and regression analysis.

Full Stack Software Engineer | Kimberly Clark, India

May 2023 – Mar 2025

- Led the implementation of 10+ REST APIs integrating SAP and Salesforce across 4 global regions, supporting 15K+ monthly API transactions for pricing workflow using Node.js and Redis.
- Owned full-stack development of an enterprise pricing application, delivering 50+ features across 4 global regions and processing 200+ pricing agreements weekly, achieving over 97% user acceptance using C# and ASP.NET Core.
- Optimized SQL Server performance across 4 databases and regions, reducing API response time from 5 to 4.4 minutes across 1000 test cases and reducing 10K record report downloads to 30 seconds.
- Migrated a legacy C# .NET codebase to ASP.NET Core and modernized it into 4 independent microservices, reducing synchronization failures and production support incidents by 32% using Docker and Kubernetes.

Associate Software Engineer | Kimberly Clark, India

Sep 2021 – Apr 2023

- Developed IoT monitoring dashboards supporting 5,000+ users across 33M+ sq. ft. in 15+ U.S. states and Canada, contributing to 105% higher visitor satisfaction and 80% lower consumable waste using Python, React and JavaScript.
- Built serverless Azure pipelines processing 1M+ daily IoT events across 2,000+ devices, cutting API response times from 40 to 20 seconds and eliminating 15 monthly complaints and work orders while saving \$68,052 annually per user.
- Implemented automated testing across 100+ workflows for 1 enterprise web and mobile application, expanding test cases from 1,200 to 1,800+ while identifying pre-release defects and regressions using Jest, pytest, and React Testing Library.
- Optimized Azure CI/CD pipelines across 4 repositories supporting 1 application, reducing deployment time from 25 to 12 minutes and enabling faster, more frequent releases through Docker caching, parallel execution, and automated quality gates.

Software Engineer Intern | Kimberly Clark, India

Jan 2021 – Aug 2021

- Designed and deployed a centralized consulting webplatform serving 3,000+ global team members and processing 75 weekly requests, reducing processing from 1 day to 4 hours while standardizing request coordination using Power Platform Tool.
- Built 15+ reusable UI components supporting 10 teams of 5 members each, reducing duplicated development effort and improving application UI consistency using Microsoft Power Apps, Power Fx, and JavaScript.
- Migrated 20+ WordPress pages to SharePoint and created 10+ onboarding guides, increasing daily page engagement from 30 to 100 views while improving information accessibility for 3,000+ global team members.

PROJECTS

Geospatial Web Platform | [Live Site](#)

| Jan 2026 – May 2026

- Engineered a GIS platform reducing geological data query time from 40 mins to 2 secs and enabling interactive visualization across 12 map layers for Delaware Mountain Group exploration using React, FastAPI, and Mapbox GL JS.
- Developed ETL pipeline consolidating 2TB+ of geological data from 30+ source files and processing 10GB+ per execution, eliminating manual consolidation and standardizing geospatial data storage using PostgreSQL, Supabase, and PostGIS.

Personal Portfolio & AI Voice Agent | [GitHub](#)

| Sep 2026

- Built a RAG-powered AI voice agent, enabling conversational Q&A grounded in portfolio and resume data, by integrating the OpenAI API with vector-based retrieval.
- Automated portfolio development and code-review workflows, reducing manual implementation and validation effort, by orchestrating specialized coding, reviewer, and conversational agents through Claude Code.

Assessing Response Consistency across Query Variants in LLM | [GitHub](#)

| Dec 2025

- Built an automated benchmarking pipeline comparing 5 OpenAI models across 10,000+ paraphrased queries, reducing processing from 3,199 to 1,530 secs and enabling faster model and prompt-consistency analysis using Python, DuckDB and Apache Arrow UDFs.
- Evaluated Chain-of-Thought prompting across 2,000 paraphrased queries and 5 OpenAI model versions, achieving 15% higher technical-reasoning scores versus 3% for general text while identifying reasoning inconsistencies through comparative prompt evaluation.

Product Review Summarization Platform | [GitHub](#)

| May 2026

- Built a distributed NLP pipeline processing 22GB of Amazon reviews, achieving 84.03% sentiment accuracy and 0.8431 weighted F1 while generating product-level intelligence on sentiment, strengths, weaknesses, and recurring complaints using Llama 3.1, BGE-M3, and BERTopic.
- Implemented Power BI dashboards analyzing review data across 12 product categories, visualizing sentiment trends, review activity, complaint rates, and semantic clusters to support product buyers and business stakeholder analysis.

PUBLICATION

Automation Leaf Disease Detection using CNN Algorithm

Springer ICTIS 2020

- Developed and trained a CNN-based image classification model on 20GB+ of apple, corn, and tomato leaf images, achieving 96% test accuracy in disease detection using Python, TensorFlow, and PyTorch.

TECHNICAL SKILLS

Programming Languages & Frameworks: Python, C#, SQL, JavaScript, TypeScript, HTML, CSS, React, Node.js, FastAPI, Jest, pytest, Playwright, Selenium, Cypress

Development Tools & Platforms: PostgreSQL, Microsoft SQL Server, MySQL, Supabase, Azure Cosmos DB, Redis, AWS, Microsoft Azure, Docker, Kubernetes, Azure DevOps, Git, GitHub, Jira, Postman, Swagger/OpenAPI

AI/ML Tools: TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face, Sentence-Transformers, LangChain, LangGraph, MCP, Jupyter Notebook, Pinecone, Claude Code